

Answer all the following questions:

1. Having the following XML document:

(3 pts)

- a. <AAA>
- b. <CCC>
- c. <BBB/>
- d. <BBB/>
- e. <BBB/>
- f. </CCC>
- g. <DDD>
- h. <BBB/>
- i. <BBB/>
- j. </DDD>
- k. <EEE>
- l. <CCC/>
- m. <DDD/>
- n. </EEE>
- o. </AAA>

-What are the results of the XPATH expressions applied to the above XML document? Indicate the number of line or lines affected in each answer

- //*[count(BBB)=2]
- //*count(*)=2]
- //*starts-with(name(),'B')]
- //*[name()='BBB']
- /AAA/BBB[1]
- /AAA/CCC/BBB[last()]

2. Write an XML Schema for a bibliography. The bibliography contains:

(5 pts)

- a. books and articles
- b. The necessary information for a book are:
 - i. Its general title
 - ii. The names of authors
 - iii. Its volumes and for each volume, the number of pages
 - iv. The general information about the edition, such as: editor, publisher, printing place, and its ISBN number.
- c. The necessary information for an article are:
 - i. Its title
 - ii. The names of authors
 - iii. Its references of publication: journal name, number of pages, publication year, and journal number.
- d. We also reserve an optional field, for each book and each item for personal conclusion.

3. We want to represent in XML a catalog of books for a library. For each book, we want to memorize the following information: number of book reference, title author(s) (at least one occurrence), editor, release date, status (available, borrowed, disappeared). In case of borrowing: borrow date, reference borrower, limit return date, and recall dates (infinite times)

For each borrower: last name, first name, address, phone num, person reference number, list of borrows, reference number for each book.

Construct several variances of an XML document containing some books and borrowers, for example by representing in variance1, all the information as elements, then, in another variance, representing some data as attributes. (3 pts)

4. Having the following XML document:

(5 pts)

```
<Company>
  <Address xsi:type="US-Address">
    <Name>Main Office</Name>
    <Street> Grosvenor Blvd </Street>
    <City> Los Angelos </City>
    <State> California </State>
    <Zip> 2921 </Zip>
  </Address>
```

```
<Division>
  <Division-Name>Sales</Division-Name>
  <Location>Washington</Location>
  <Person Manager="true" Degree="MBA">
    <First> Allison</First>
    <Last> Anderson</Last>
    <PhoneExt>111</PhoneExt>
    <Email>Anderson@hotmail.com</Email>
  </Person>
  <Person Manager="false" Degree="BA">
    <First> Bruce</First>
    <Last> Burrwinkle</Last>
    <PhoneExt>222</PhoneExt>
    <Email> Burrwinkle@hotmail.com</Email>
  </Person>
</Division>
<Division>
  .....
  .....
  .....
</Division>
</Company>
- Indicate, in the case where it is possible, how the following constraints can be represented via a DTD and an XML Schema:
  • The value of "Degree" is MBA or BA
  • The Email address has the following form: "string.string@string"
  • US-Address is an extension of "Address" to which we add the elements "state" and "zip"
  • The element "zip" has a value between 1000 and 4000
  • The persons, in the XML instances, have to be sorted in alphabetical order.
```

5. Write an XSL document for the following XML document that gives the output shown below (Figure 1). (5 pts)

```
<?xml version="1.0"?>
<?xml:stylesheet type="text/xsl" href="games.xsl"?>
<sports>
<game id="567">
<name>Cricket</name>
<paragraph>More popular among commonwealth nations</paragraph>
</game>
<game id="568">
<name>Baseball</name>
<paragraph>More popular in America</paragraph>
</game>
<game id="569">
<name>Soccer</name>
<paragraph>More popular in the world</paragraph>
</game>
<game id="570">
<name>ping Pong</name>
<paragraph>More popular in the world</paragraph>
</game>
<game id="571">
<name>Hockey</name>
<paragraph>More popular in USA</paragraph>
</game>
```

NB: The background color of the table is gray. We want to display ONLY the first three elements "game" in the table. (Figure 1). An extra row will be added to the output (Figure 1) to show the position number of each "Id"

Figure 1

Node Position	Id	Name	Paragraph
1	567	Cricket	More popular among commonwealth nations
2	568	Baseball	More popular in America
3	569	Soccer	More popular in the world